

Basic Stuff

Translational Inertia Element (Mass):

$$m\ddot{x}(t) = \Sigma F = F_m(t)$$

Translational Stiffness Element

(Spring):

$$F_k = k[x_2(t) - x_1(t)]$$

Translational Damping Element

(Damper):

$$F_b(t) = b[\dot{x}_2(t) - \dot{x}_1(t)]$$

Rotational Inertia Element:

$$J\ddot{\theta} = T_j(t), \theta_1$$

Rotational Stiffness Element

(Spring):

$$T_k(t) = k[\theta_2(t) - \theta_1(t)]$$

Rotational Damping Element

(Damper):

$$T_b(t) = b[\dot{\theta}(t) - \dot{\theta}_1(t)]$$

Fluid Systems

Reynolds Number:

$$N_R = \frac{\rho v D}{\mu} = \frac{v D}{\nu}$$

*If $N_R < 2000$, the flow is Laminar
If $N_R > 4000$, the flow is Turbulent
If $2000 < N_R < 4000$, Critical Zone*

$$N_R = \frac{\rho \left(\frac{kg}{m^3} \cdot \frac{slugs}{ft^3} \right) v \left(\frac{m}{s} \cdot \frac{ft}{s} \right) D(m, ft)}{\mu \left(\frac{N \cdot s}{m^2} \cdot \frac{lb \cdot s}{ft^2} \right)} = \frac{v \left(\frac{m}{s} \cdot \frac{ft}{s} \right) D(m, ft)}{\nu \left(\frac{m^2}{s} \cdot \frac{ft^2}{s} \right)}$$

ρ (rho) = the density
 μ (mu) = the absolute viscosity
 ν (nu) = the kinematic viscosity
 v = fluid velocity
 D = pipe diameter

The Continuity Equation:

$$(1) \dot{m} = \Sigma w_{in} - \Sigma w_{out}$$

$$(2) \dot{m} = \rho \dot{V} = \rho A \dot{h}$$

m	Fluid mass
v	Fluid volume
ρ	Fluid density
w_in	Mass flow rate into system
w_out	Mass flow rate out system

Note: $\dot{m} = 0$ if steady flow thru CV

Fluid Resistance:

$$(3) w = \frac{1}{R} (p_1 - p_2) \left(\frac{1}{A} \right)$$

$$(4) w = pQ$$

w	Mass flow rate
Q	Volume flow rate
p	Fluid density
p	Pressures
R, a	Constant (restriction type)

Fluid Capacitance (C):

$$(5) F = pA$$

$$(6) p = pgh + P_{atm}$$

$$(7) \dot{p} = \frac{1}{C} (Q_{in} - Q_{out})$$

$$(8) C = \frac{A}{\rho g}$$



F	Force
P	Pressure in fluid
A	Area on which fluid acts
P	Fluid density
G	Gravitational acceleration
H	Fluid height
p_atm	Atmospheric pressure
C	Fluid capacitance
Q	Volume flow rate

Modeling Hydromechanical System:

$$(9) \dot{p} = \frac{\beta}{V} (Q_{in} - \dot{V})$$

$$(10) V = V_o + Ax, \dot{V} = A\dot{x}$$

P	Pressure in the fluid
V	Volume
Q	Volume flow rate
B	Fluid-bulk modulus
Vo	Volume out
A	Piston cross sectional area
X	Displacement of piston

The β for air is 1.40×10^5

The β for water is 2.25×10^9

Thermal Systems

Net Rate of Heat Energy Stored:

$$\dot{\xi}_h = \Sigma h_{in} - \Sigma h_{out} + \Sigma q_{in} - \Sigma q_{out}$$

H	Time rate of enthalpy due to mass transfer across boundary
Q	Heat flow rates across boundary

Thermal Resistance (11,12,13):

$$q = \frac{1}{R} \Delta T$$

$$R = \frac{x}{kA} \text{ (Conduction Heat Transf.)}$$

$$R = \frac{1}{hA} \text{ (Convection Heat Transf.)}$$

q	Heat flow rate (J/s or W)
ΔT	Temp difference, deg K
R	Thermal resistance, deg K-s/J
A	Area
H	Convection coefficient
k	Thermal conductivity coef

Note: Perfect insulator is $q = 0$

Thermal Capacitance (C)(J/deg K):

$$(14) C = mc_p$$

$$(15) C = mc_v$$

M	Pressure in fluid
c_p	Area on which fluid acts
c_v	Fluid density

Modeling Thermal System:

$$(16) C\dot{T} = \Sigma q_{in} - \Sigma q_{out}$$

$$(17) q = \frac{1}{R} (T_1 - T_2)$$

q	Heat-energy flow
R	Thermal resistance
T	Temperature
C	Thermal capacity

Thermal Material Properties (18, 19):

$$\frac{1}{R} = \frac{kA}{l} \text{ (k and R relationship)}$$

$$q = wC_v(T_1 - T_2)$$

(Warm to Cool / Cool to Warm Heat Flow)

$$(20) w = KA$$

C	Thermal capacity
m	mass
C_v	Specific heat at constant volume
R	Thermal resistance
k	Thermal conductivity
A	Cross-sectional Area
l	Length of heat-flow path
w	Mass flow rate of fluid
K	Flow coefficient

Fluid (2-Tank):

Write eq.1, Write eq.2 for each mass, Relate eq.2=eq.1, define which Ws used for eq.1, Define the w outputs as eq.3., substitute em into into eq.2 =eq.1, then isolate h1,h2.

Fluid (Hydraulic Piston):

Write eq.9, Write eq.10, then sub Vdot into eq.9. Make 2D FBD for I/O, use BASIC to find EoM.

Rearrange final ans to Inputs = Outputs. (Btw, $Q_{in}=PA$ (Piston Area))

Thermal (Block to Block)

Write eq.16 for each mass, define which q's, sub the q's as eq.17s, rearrange terms to isolate Tdot1s

Thermal & Fluid Answer Formatting

$$\dot{T}_2 = \frac{1}{C_2 R_x} (T_1 - T_2) - \frac{1}{C_2 R_2} (T_2 - T_o)$$

$$m \ddot{x} = \Sigma F \quad m \ddot{x} = PA - P_{atm}A - kx - b\dot{x} - F_L$$

→

$$\dot{T}_2 = \left(\frac{1}{C_2 R_2} \right) T_o + \left(\frac{1}{C_2 R_x} \right) T_1 - \left(\frac{1}{C_2 R_x} + \frac{1}{C_2 R_2} \right) T_2$$

$$m \ddot{x} + b \dot{x} + k x = (P - P_{atm}) A - F_L$$